

Exercise #7

27. February 2023

Problem 1. (PDEs)

- a) Check that $u = \frac{1}{16\pi^4 - 4\pi^2} \sin(2\pi x)$ solves the following equation on $[0, 1]$,

$$\partial_x^4 u + \partial_x^2 u = \sin(2\pi x),$$

where $u(0) = u(1) = 0$.

- b) Determine the constant $\alpha \in \mathbb{R}$ so that $u = \frac{1}{16\pi^4 - 4\pi^2} \sin(2\pi x) + \alpha \sin(4\pi x)$ solves,

$$\partial_x^4 u + \partial_x^2 u = \sin(2\pi x) + \sin(4\pi x), \quad u(0) = u(1) = 0.$$

Problem 2. (Superposition)

In this problem we will consider the heat equation on $[0, 1]$:

$$\partial_t u - \partial_x^2 u = f(x). \tag{1}$$

- a) Consider (1) when $f(x) = 0$. Show that,

$$u_1 = e^{-4\pi^2 t} \cos(2\pi x), \quad \text{and} \quad u_2 = e^{-16\pi^2 t} \cos(4\pi x),$$

solves (1) in this case.

b) Argue that $u_1 + u_2$ is also a solution to (1) in this case.

c) Set $f(x) = \sin(2\pi x)$. Show that,

$$u_3 = \frac{1}{4\pi^2}(1 - e^{-4\pi^2 t}) \sin(2\pi x),$$

is a solution to (1) in this case. Is $u_3 + u_3$ a solution to (1) in this case?

d) We now impose boundary conditions and an initial condition on (1):

$$u(0, x) = g(x), \quad u(t, 0) = u(t, 1) = 0.$$

Set $f(x) = \sin(2\pi x)$ and $g(x) = 0$, and check that u_3 (from point c) solves (1) with these boundary conditions, and the initial condition.

e) Now, set $f(x) = 0$ and $g(x) = \sin(4\pi x)$, and denote the solution to (1) by u_4 . Check that,

$$u_4 = e^{-16\pi^2 t} \sin(4\pi x).$$

Problem 3. (Classification of PDEs)

Classify the following PDEs into linear/nonlinear, homogeneous/inhomogeneous:

a) One dimensional heat equation,

$$\frac{\partial}{\partial t} u = \partial_x^2 u$$

b) One dimensional heat equation on $[0, 1]$ with varying diffusion,

$$\frac{\partial}{\partial t} u = x(x - 1) \partial_x^2 u$$

c) The Laplace equation in polar coordinates ($u = u(r, \theta)$):

$$\frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r} \frac{\partial^2 u}{\partial \theta^2} = 0.$$

d) The Poisson equation describing torsion of prismatic bars:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + 3 = 0.$$

e) The convection-reaction equation, with a non-zero reaction exponent $n \in \mathbb{N}$:

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} + u^n = 0$$

f) The bi-harmonic equation in two dimensions:

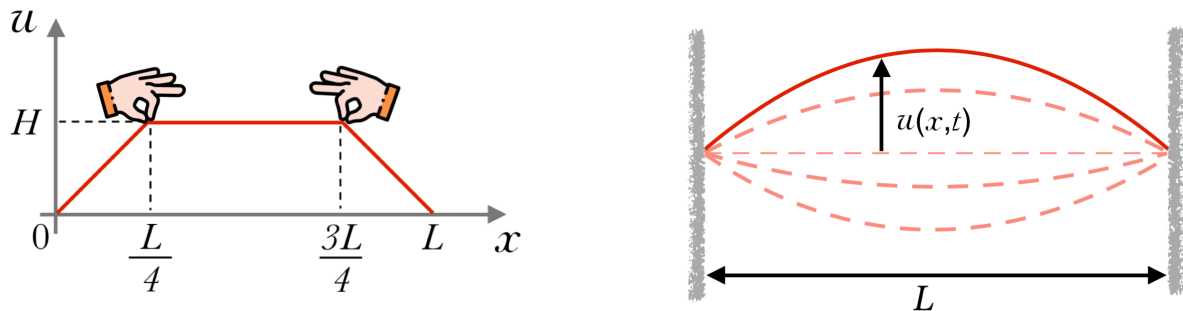
$$\frac{\partial^4 u}{\partial x^4} + 8 \frac{\partial^4 u}{\partial x^2 \partial y^2} + \frac{\partial^4 u}{\partial y^4} = 0.$$

Problem 4. (Wave equation)

The small transversal displacements $u(x, t)$ of a stretched string satisfy the wave equation:

$$\frac{\partial^2 u}{\partial t^2} = \frac{FL}{m} \frac{\partial^2 u}{\partial x^2}, \quad (2)$$

where L and m are the string's length and mass, respectively, and F is the tension pulling the string (all three of them constant). With the string fixed on both sides, we get the boundary conditions $u(0, t) \equiv u(L, t) \equiv 0$. Let's say we grab the string at two points $x = L/4$ and $x = 3L/4$, and pull it up until reaching a height H . Starting from this trapezoidal shape at $t = 0$, the string is released and starts vibrating:



This initial condition for $u(x, t)$ can be written either in three separate linear parts, or also more compactly as

$$u(x, 0) = \left(\frac{1}{2} + \left| \frac{2x}{L} - 1 \right| - 1 \right) H. \quad (3)$$

Moreover, since the string is simply *released*, the velocity $\frac{\partial u}{\partial t}$ will be zero at $t = 0$.

- Determine the wave speed c in terms of the given constants F , L and m .
- Determine the vibration period in terms of F , m and L .
- Using separation of variables, find a general solution $u(x, t)$ for Eq. (2) fulfilling the boundary conditions.
- Finally, considering $L = 1$, apply the initial conditions to find the solution $u(x, t)$ of this initial-boundary value problem
Hint: if you want, you can use some symbolic integration tool to evaluate the necessary integrals.

The next exercise is optional and should not be handed in!

Problem 5. (Damped wave equation)

One way to include dissipation in wave modelling is via linear damping. In that case, the wave equation becomes

$$\frac{\partial^2 u}{\partial t^2} + 2\gamma \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2},$$

in which c is the wave speed and $\gamma > 0$ is the damping parameter. As for the standard wave equation, we need boundary and initial conditions. Considering the following ones

$$\begin{aligned} u(0, t) &= 0, \\ u(L, t) &= 0, \\ u(x, 0) &= f(x), \\ \left. \frac{\partial u}{\partial t} \right|_{t=0} &= 0 \end{aligned}$$

and assuming $\gamma < c\pi/L$, use the separation of variables technique to show that the solution to this problem is

$$u(x, t) = \sum_{n=1}^{\infty} C_n e^{-\gamma t} \sin(\lambda_n x) \cos\left(t\sqrt{(c\lambda_n)^2 - \gamma^2}\right), \text{ with } \lambda_n := \frac{n\pi}{L}, \quad n \in \mathbb{N},$$

and determine the constants C_n in terms of $f(x)$.